



COMPUTER SECURITY

Lecture 2: Cryptography

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Why Encryption?

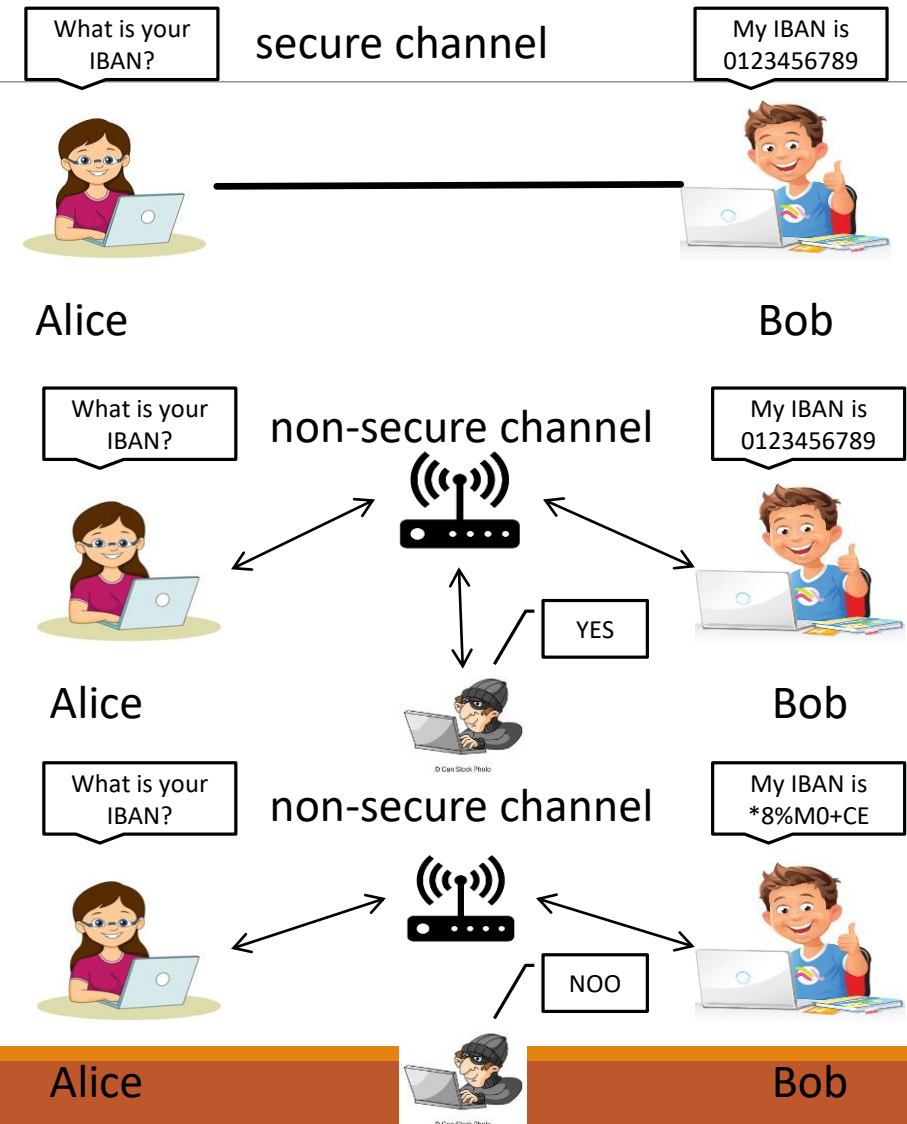
Assuming that Alice and Bob are communicating with each other:

Alice would like to have Bob's IBAN number.

Three scenarios might occur:

1. Communicating over a secure channel
2. Communicating over a non-secure channel in plaintext.
3. Communicating over a non-secure channel in encrypted text.

Next week we will go over encryption in greater details.



Problems addressed by encryption

Assuming that, a **sender** (S) sends a message to a **recipient** (R) over a **transmission medium** (T).

If an **outsider** (O) wants to access the message, we call him intruder (or interceptor).

The outsider might take one (or more) of the following actions:

1. Block the message from reaching R. (Affecting availability)
2. Intercept the message by reading or listening. (affecting confidentiality)
3. Modify the message. (Affecting integrity)
4. Fabricate the message. (affecting the integrity)

Some Basic Terminology

- **plaintext** - original message
- **ciphertext** - coded message
- **cipher** - algorithm for transforming plaintext to ciphertext
- **key** - info used in cipher known only to sender/receiver
- **encipher (encrypt)** - converting plaintext to ciphertext
- **decipher (decrypt)** - recovering plaintext from ciphertext
- **cryptography** - study of encryption principles/methods
- **cryptanalysis (codebreaking)** - study of principles/ methods of deciphering ciphertext *without* knowing key
- **cryptology** - field of both cryptography and cryptanalysis

Terminology

Encryption (E) (encoding -enciphering): The process of encoding a message so that its meaning is not obvious.

Decryption(D)(decoding - deciphering): Is the reverse process.

Cryptosystem: A system that encrypts/decrypts.

Plaintext (P): The original form of a message.

Cipher text (C): the encrypted form of a message.

NOTATION

Processing a plaintext (P) using the encryption rule (E), results on cipher text (C), and denoted by:

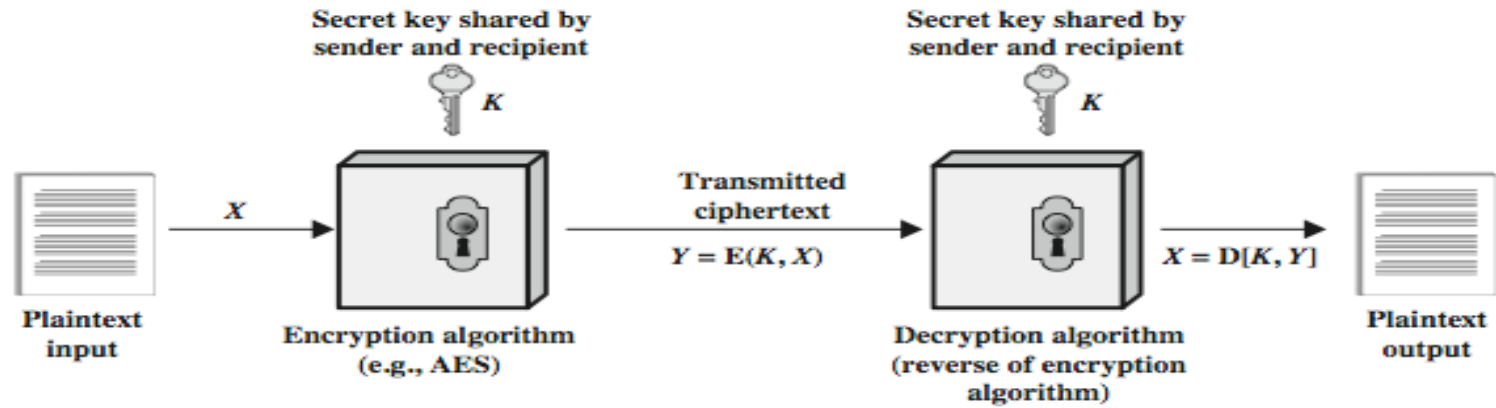
$$C = E (P)$$

The reverse process is denoted by:

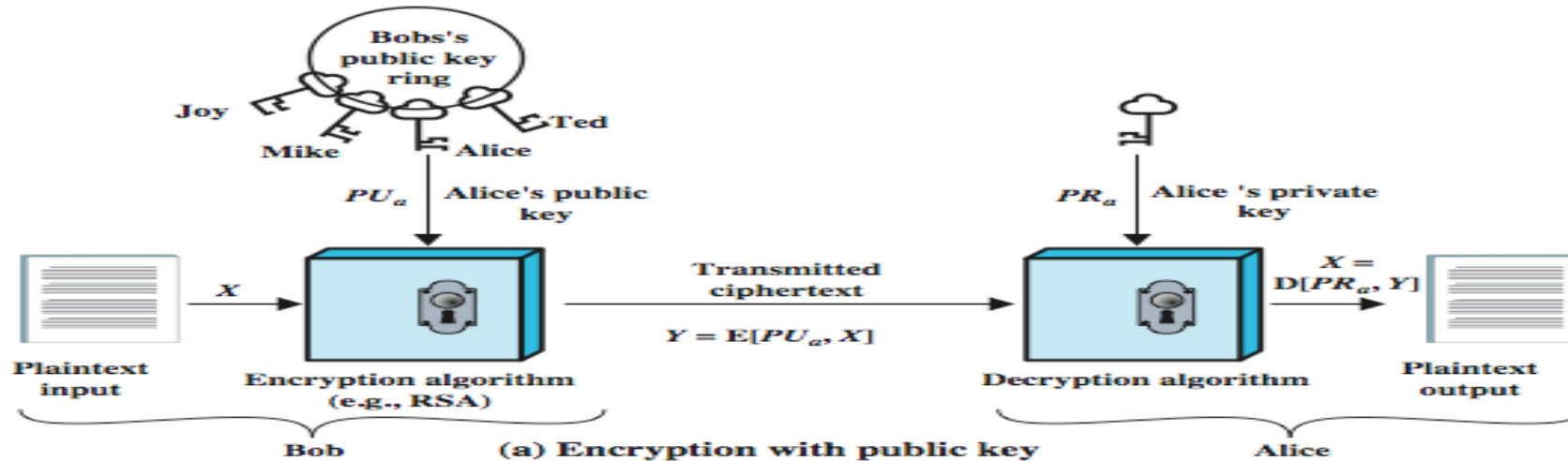
$$P = D (C)$$

Question:

What is $D(E(P))$?



Symmetric Cipher Model



Asymmetric Cipher Model

Requirements for secure use of symmetric encryption:

➤ two requirements

- a strong encryption algorithm
- a secret key known only to sender / receiver

➤ mathematically have:

$$Y = E(K, X) = E_K(X) = \{X\}_K$$

$$X = D(K, Y) = D_K(Y)$$

➤ assume encryption algorithm is known

- Kerckhoff's Principle: security in secrecy of key alone, not in obscurity of the encryption algorithm

➤ implies a secure channel to distribute key

- Central problem in symmetric cryptography

Cryptography

➤ can characterize cryptographic system by:

- type of encryption operations used
 - substitution
 - transposition
 - product
- number of keys used
 - single-key or private
 - two-key or public
- way in which plaintext is processed
 - block
 - stream

Cryptanalysis

- objective to recover key not just message
- general approaches:
 - cryptanalytic attack
 - brute-force attack
- if either succeed all key use compromised

Cryptanalytic Attacks

➤ ciphertext only

- only know algorithm & ciphertext, is statistical, can identify plaintext

➤ known plaintext

- know/suspect plaintext & ciphertext

➤ chosen plaintext

- select plaintext and obtain ciphertext

➤ chosen ciphertext

- select ciphertext and obtain plaintext

➤ chosen text

- select plaintext or ciphertext to en/decrypt

Cipher Strength

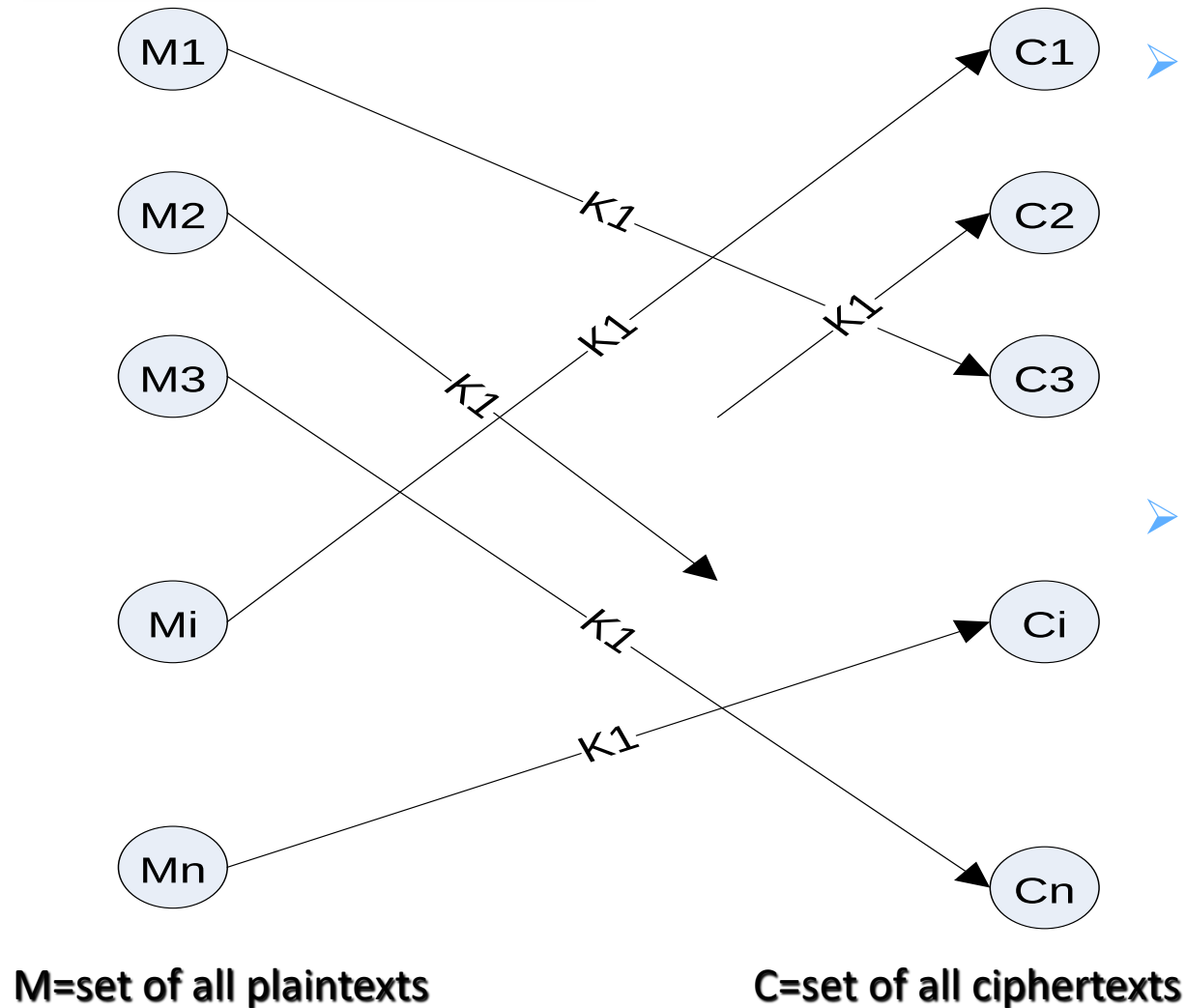
➤ **unconditional security**

- no matter how much computer power or time is available, the cipher cannot be broken since the ciphertext provides insufficient information to uniquely determine the corresponding plaintext

➤ **computational security**

- given limited computing resources (e.g. time needed for calculations is greater than age of universe), the cipher cannot be broken

Encryption Mapping (1)



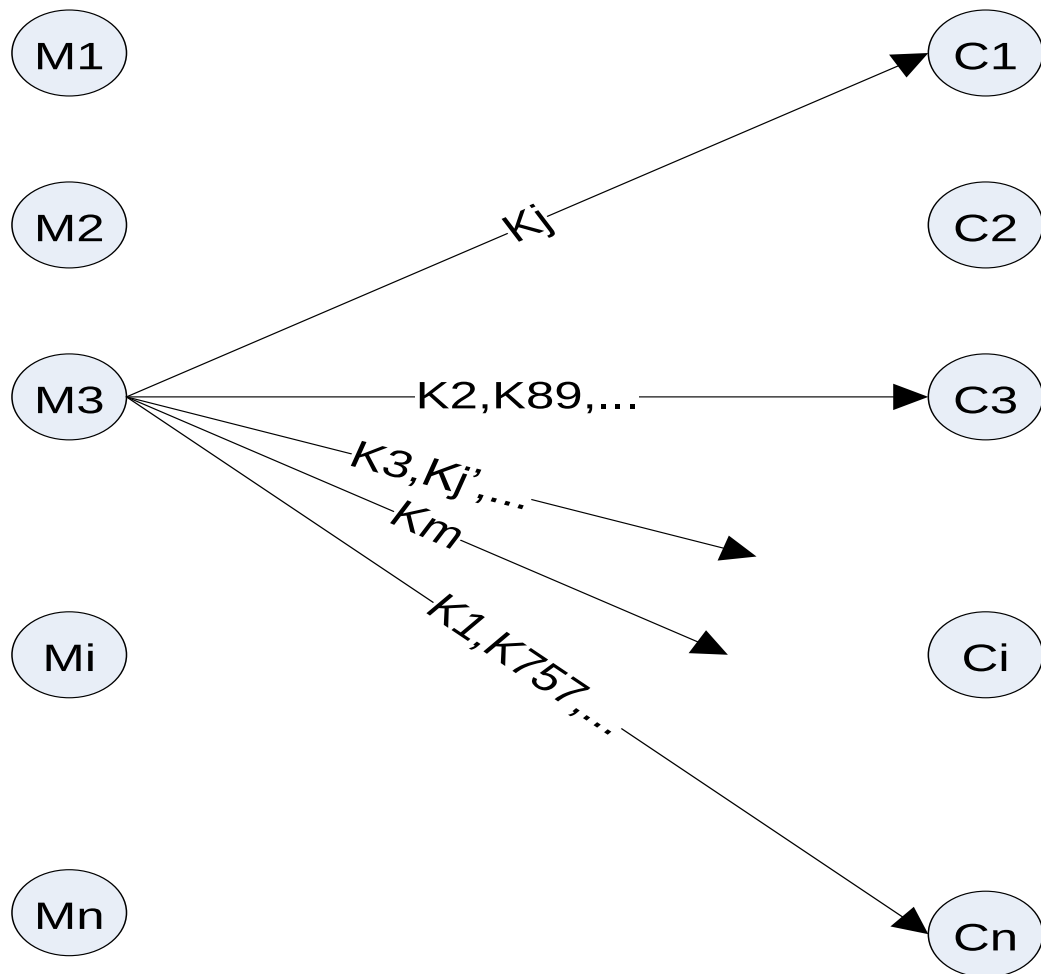
➤ A given key (k)

- Maps any message M_i to some ciphertext $E(k, M_i)$
- Ciphertext image of M_i is unique to M_i under k
- Plaintext pre-image of C_i is unique to C_i under k

➤ Notation

- key k and M_i in M , $\exists! C_j$ in C such that $E(k, M_i) = C_j$
- key k and ciphertext C_i in C , $\exists! M_j$ in M such that $E(k, M_j) = C_i$
- $E_k(\cdot)$ is “one-to-one” (injective)
- If $|M|=|C|$ it is also “onto” (surjective), and hence bijective.

Encryption Mappings (2)



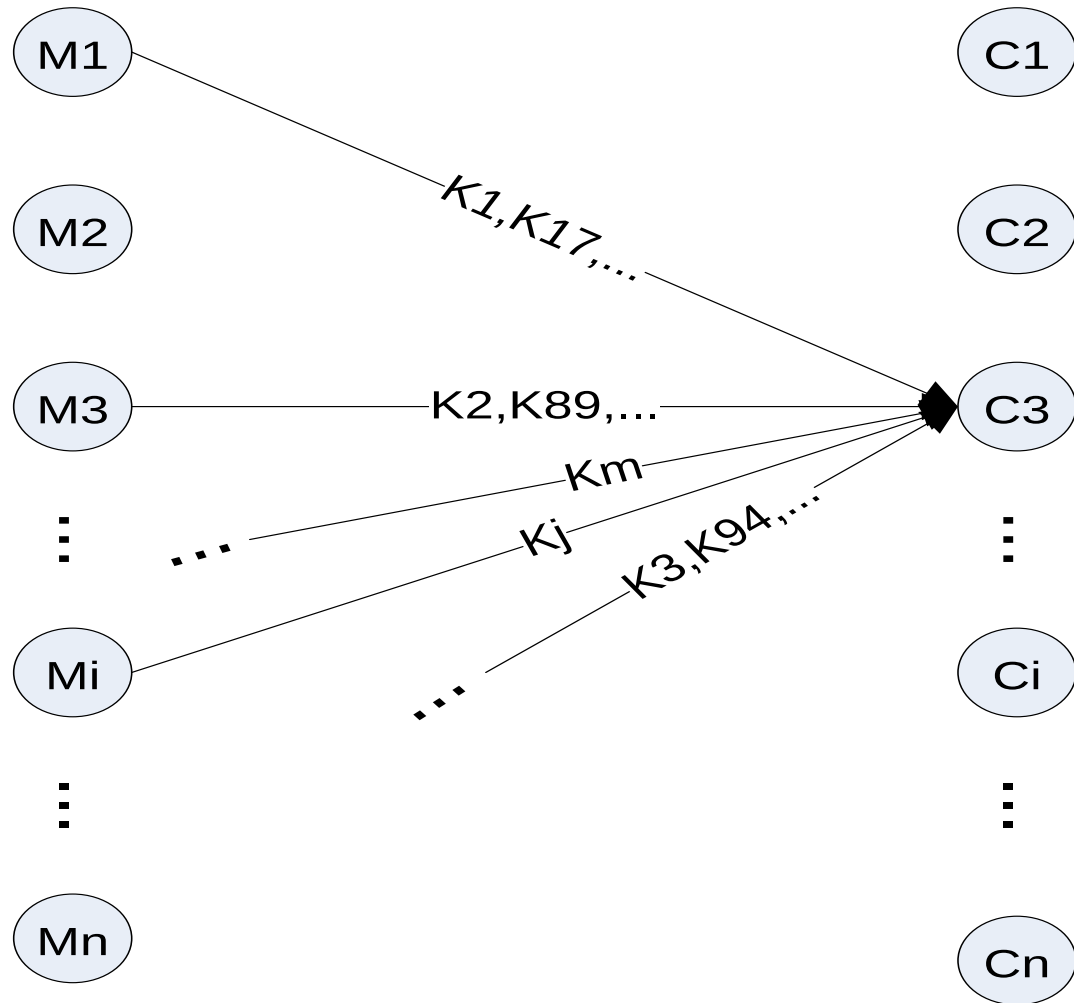
➤ A given plaintext (M_i)

- M_i is mapped to *some* ciphertext $E(K, M_i)$ by every key k
- Different keys may map M_i to the same ciphertext
- There may be some ciphertexts to which M_i is never mapped by any key

➤ Notation

- key k and M_i in M , $\exists!$ ciphertext C_j in C such that $E(k, M_i) = C_j$
- It is possible that there are keys k and k' such that $E(k, M_i) = E(k', M_i)$
- There may be some ciphertext C_j for which $\exists$ key k such that $E(k, M_i) = C_j$

Encryption Mappings (3)



➤ A ciphertext (C_i)

- Has a unique plaintext pre-image under each k
- May have two keys that map the same plaintext to it
- There may be some plaintext M_j such that no key maps M_j to C_i

➤ Notation

- key k and ciphertext C_i in C , $\exists!$ M_j in M such that $E(k, M_j) = C_i$
- There may exist keys k, k' and plaintext M_j such that $E(k, M_j) = E(k', M_j) = C_i$
- There may exist plaintext M_j such that $\exists$ key k such that $E(k, M_j) = C_i$

Encryption Mappings (4)

- Under what conditions will there always be some key that maps some plaintext to a given ciphertext?
- If for an intercepted ciphertext c_j , there is some plaintext m_i for which there does not exist any key k that maps m_i to c_j , then the attacker has learned something
- If the attacker has ciphertext c_j and known plaintext m_i , then many keys may be eliminated

Brute Force Search

- always possible to simply try every key
- most basic attack, exponential in key length
- assume either know / recognise plaintext

Key Size (bits)	Number of Alternative Keys	Time required at 1 decryption/ μ s	Time required at 10^6 decryptions/ μ s
32	$2^{32} = 4.3 \times 10^9$	$2^{31} \mu$ s = 35.8 minutes	2.15 milliseconds
56	$2^{56} = 7.2 \times 10^{16}$	$2^{55} \mu$ s = 1142 years	10.01 hours
128	$2^{128} = 3.4 \times 10^{38}$	$2^{127} \mu$ s = 5.4×10^{24} years	5.4×10^{18} years
168	$2^{168} = 3.7 \times 10^{50}$	$2^{167} \mu$ s = 5.9×10^{36} years	5.9×10^{30} years
26 characters (permutation)	$26! = 4 \times 10^{26}$	$2 \times 10^{26} \mu$ s = 6.4×10^{12} years	6.4×10^6 years

Classical Substitution Ciphers

- where letters of plaintext are replaced by other letters or by numbers or symbols
- or if plaintext is viewed as a sequence of bits, then substitution involves replacing plaintext bit patterns with ciphertext bit patterns

Caesar cipher

Caesar cipher is one of the oldest encryption algorithms.

It was created by Julius Caesar (100 – 44 BC).

The idea is simple:

1. Index every character from 0 to 25
2. When encrypting, shift every character by 3 (revise it when decrypting)

Caesar agreed with his people to use the encryption key = 3 ($K = 3$).



Caesar cipher

Example: Encrypt the word (KEY) using Caesar cipher ?

The corresponding index (i) for each letter in the plaintext is { 10, 4, 24 } respectively.

The mathematical formula is : $(i + 3) \bmod 26$.

The corresponding index (i) for each letter in the cipher text is { 13, 7, 1 } respectively.

The corresponding cipher text is (NHB).

0	1	2	3	4	5	6	7	8	9	10	11	12
A	B	C	D	E	F	G	H	I	J	K	L	M
13	14	15	16	17	18	19	20	21	22	23	24	25
N	O	P	Q	R	S	T	U	V	W	X	Y	Z

SHIFT CIPHER

Caesar cipher is a special case of the shift cipher when the key $K = 3$.

Shift cipher is a generalized formula and the value varies between 0 to 26.

Shift cipher is an example of a symmetric cryptosystem.

EX:

meet me after the toga party
PHHW PH DIWHU WKH WRJD SDUWB



Playfair Cipher

- not even the large number of keys in a monoalphabetic cipher provides security
- one approach to improving security was to encrypt multiple letters
- the **Playfair Cipher** is an example
- invented by Charles Wheatstone in 1854, but named after his friend Baron Playfair

Playfair Key Matrix

- a 5X5 matrix of letters based on a keyword
- fill in letters of keyword (sans duplicates)
- fill rest of matrix with other letters
- eg. using the keyword MONARCHY

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Playfair Encrypting and Decrypting

- plaintext is encrypted two letters at a time
 1. if a pair is a repeated letter, insert filler like 'X'
 2. if both letters fall in the same row, replace each with letter to right (wrapping back to start from end)
 3. if both letters fall in the same column, replace each with the letter below it (wrapping to top from bottom)
 4. otherwise each letter is replaced by the letter in the same row and in the column of the other letter of the pair

Playfair Example

- Message = Move forward
- Plaintext = mo ve fo rw ar dx
- Here x is just a filler, message is padded and segmented
- **mo -> ON; ve -> UF; fo -> PH, etc.**
- Ciphertext = ON UF PH NZ RM BZ

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Transposition Ciphers

- now consider classical **transposition** or **permutation** ciphers
- these hide the message by rearranging the letter order
- without altering the actual letters used
- can recognise these since have the same frequency distribution as the original text

Rail Fence cipher

- write message letters out diagonally over a number of rows
- then read off cipher row by row
- eg. write message out as:
m e m a t r h t g p r y
e t e f e t e o a a t
- giving ciphertext
MEMATRHTGPRYETEFETEOAAT

Row Transposition Ciphers

- is a more complex transposition
- write letters of message out in rows over a specified number of columns
- then reorder the columns according to some key before reading off the rows

Key: 4312567 p: attack postponed until two am

Column Out 4 3 1 2 5 6 7

Plaintext: a t t a c k p
 o s t p o n e
 d u n t i l t
 w o a m x y z

Ciphertext: TTNAAPTMTSUOAODWCOIXKNLYPETZ

Block Transposition Ciphers

- arbitrary block transposition may be used
- specify permutation on block
- repeat for each block of plaintext

Key: 4931285607

Plaintext: attackpost poneduntil twoamxyzab

Ciphertext: CTATTSKPAO DLEONIDUPT MBAWOAXYTZ ???



Product Ciphers

- ciphers using substitutions or transpositions are not secure because of language characteristics
- hence consider using several ciphers in succession to make harder, but:
 - two substitutions make a more complex substitution
 - two transpositions make more complex transposition
 - but a substitution followed by a transposition makes a new much harder cipher
- this is bridge from classical to modern ciphers

Data Encryption standard (DES)

It was developed in 1970 by IBM.

DES is a block ciphering algorithm.

DES is a symmetric encryption algorithm.

DES is complex encryption algorithm that uses: substitution and transposition.

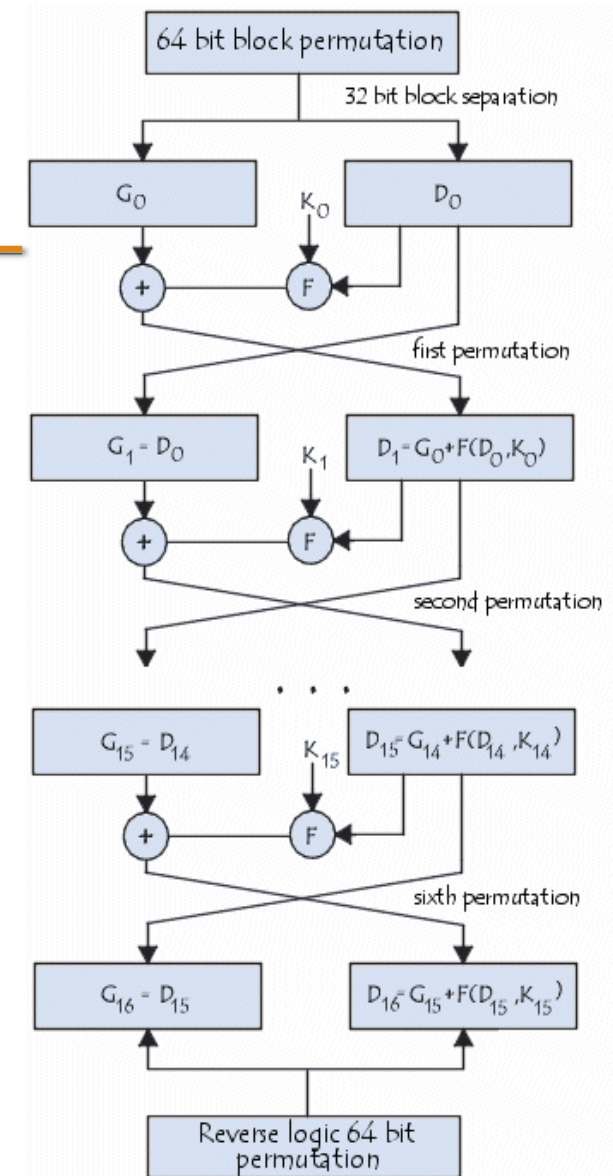
DES starts by chopping each message into 64bits (8 bytes).

The key is (also) 64 bits + 8 bits for check digits = 56 bits.

16 iterations are involved.

In 1977, Diffie and Hellman argued that 56-bits key is short.

In 1997, 3500 of parallel PC was able to break a DES cipher in 4 months.



Double DES

In 1998, a US\$ 200,000 “DES cracker” machine was invented to crack any DES in 4 days.

Later, DES is deciphered in few hours.

Double Key DES was invented later followed by Triple DES.

Double DES uses the following method:

$$E(k_2, E(k_1, m))$$

In theory, double DES multiply the difficulty of breaking the code.

Merkle and Hellman double DES proved that using double 56bits key results in similar result as if we were using single 57bits key.

Therefore, DES adds no security.

Triple DES

The cipher texted is produced using 3DES as follows:

$$E(k_3, E(k_2, E(k_1, m)))$$

The key produced is roughly equivalent to 112-bit key.

Two-Key 3DES is invented using the following formula:

$$E(k_1, D(k_2, E(k_1, m)))$$

The key produced is roughly equivalent to 80-bit key.

A comparison between all DES is available in the book in page 97

Breaking double DES requires 2000 years

Breaking a 3DES is significantly longer.

In 1995, Advanced Encryption Slandered (AES) was introduced.

In 2001, AES became the standard encryption for all federal government.

Steganography

- an alternative to encryption
- hides existence of message
 - using only a subset of letters/words in a longer message marked in some way
 - using invisible ink
 - hiding in LSB in graphic image or sound file
 - hide in “noise”
- has drawbacks
 - high overhead to hide relatively few info bits
- advantage is can obscure encryption use

THANK YOU

